Report: Location Logger

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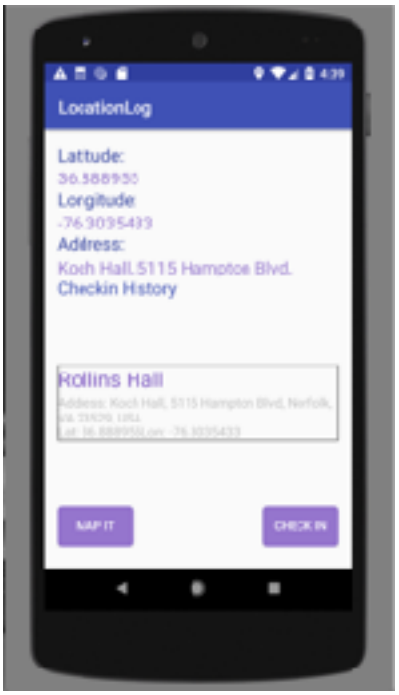
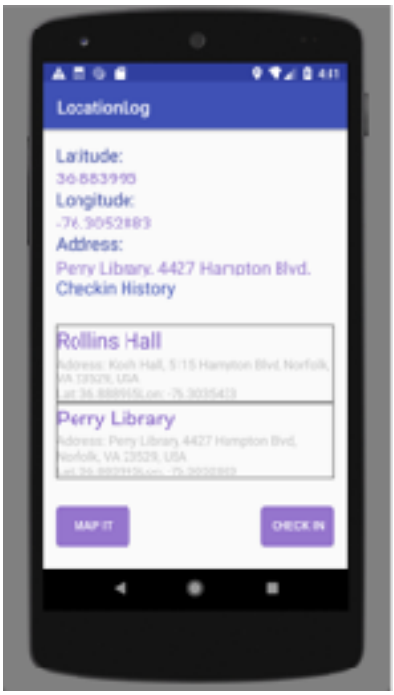
1. Most of part one of the assignment takes place in the initial launch screen.
   A. Upon launch, the last known location is obtained using the getFusedLocationProviderClient and a function called startLocationStuff. The Lat Lon are displayed at the top of the Main Activity.
   B. Once the location has been found, an IntentService is launched which fetches the address in the background.
   C. mLocationCallback is defined in onCreate(). It passes the location to a function called setData() which calls the intentService for the address and handles updating the current location. It also updates the current Checkin Object.
   D. Checkin appears bottom right of the MainActivity. It launches an alert Dialog to handle each checkin and name. Checkins are stored int checkinDB using the current CHECKIN object. The Database consists of two tables: checkins and places. The list of checkins appears below Checkin History in the MainActivity.
   E. When the alert dialog pops up, it gives the user the opportunity to assign a name to the checkin. This is saved in the places table while the checkin holds the PID of the associated place in its table.
   F. When a location is updated, the location is compared to the places table to see if it is within 30 meters of a known location. This occurs in the setData() function, where it calls checkForName() and uses Location.distanceto() to check if its nearby. When the user clicks checkin, the probable checkin name is offered as a name for this location. The user can accept the name, or decline it and enter a new name. This occurs in the alert dialog. The displayed checkin list uses the arrayList checkinList and uses the PID associated with each checkin to fetch place names for each checkin from the places table. Each place is stored only once, and each checkin is stored once.
2. On the bottom left of the MainActivity there is a Map It button which takes you to the MapsActivity.
   A. The current Location is obtained using locationManager.getLastKnownLocation(bestProvider). Using the returned location, the camera is moved in a function called locationUpdate(). Buttons and map controls are setup using UISettings in the onMapReady() function.
   B. Checkins and Places are plotted on the map in the function PopulateMap() which is called in OnMapReady() and whenever the list of places is updated.

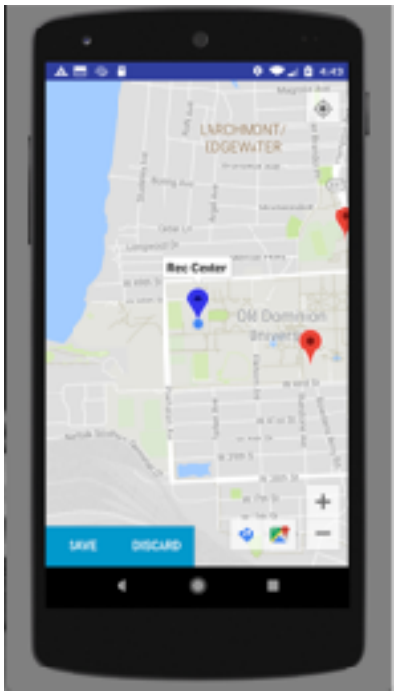
C. To add a marker, there is a button in the bottom left. This starts a dialog which allows the user to add a name. I didn't use the on map click to do this because the onMapClick wasn't behaving in the emulator. It kept creating new points when I was trying to drag other markers or interact with the UI controls. After the Dialogue box clears, a Blue Marker appears on the map at the users current location. They are then able to drag the marker around the map to place it. Once the marker is placed, the user can click save, to add the place to the places table in the database, or discard to remove the marker. Once save is clicked, the Marker turns red, and the save and discard buttons are replaced by the add button.

3. I was unable to complete the final portion of the assignment. I do not have an android phone and using the emulator does not enable you to log network responsiveness or location accuracy as there is no network.

Checkin at Koch Hall (accidentally named it Rollins) and Perry Library.

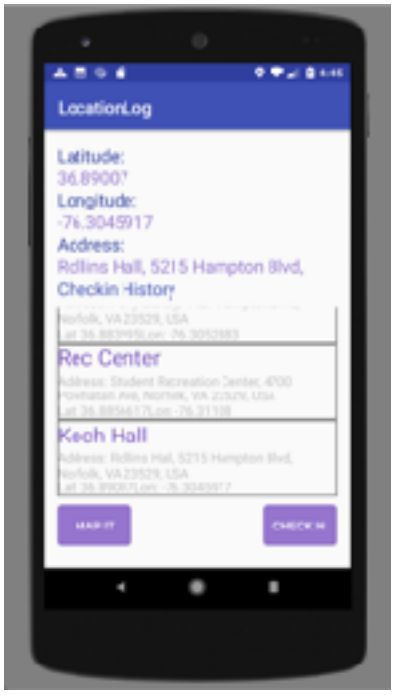




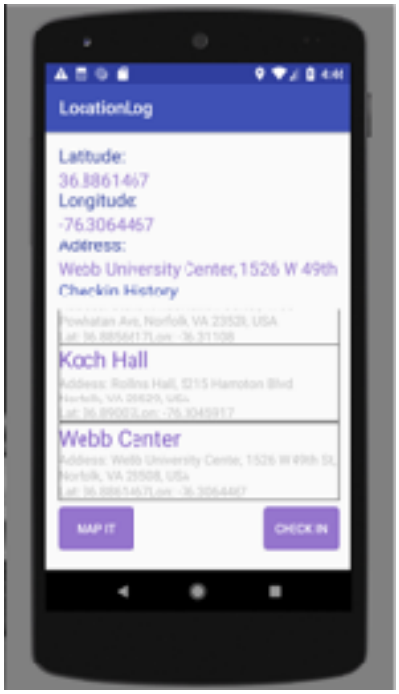


Pin added for the Rec Center

After adding the Pin, returned to the MainActivity and checked into the Rec Center, which was suggested as a place because the Rec Center was added to the places table in the MapsActivity.



Checked into Rollins Hall (named it Koch just because).



Checked into the Webb Center.